

Determinant	Definition/Operationalization	Connection to Health Issue	Connection to Determinants	SEF Level/Justification	Importance
Level of Antimicrobial Resistant Bacteria in Meat	The level of antimicrobial resistant bacteria in meat indicates the number of harmful bacteria present in animal products who display resistance to antibiotics. An example of this is E.coli with ESBL resistant genes. ¹ One method of measurement is culturing bacteria and isolating harmful Extended-Spectrum Beta-Lactamases to determine the risk associated with the bacteria, then treating the cultures antibiotics to measure resistance to each one. ¹	Antibiotic use in meat animals leads to antibiotic residues, which are classified as subtherapeutic doses, which can lead to bacterial adaptation within the organisms. ² The bacteria is then transmitted to humans through consumption, known as farm to fork transmission. ³ This in turn increases the risk of antimicrobial resistant bacteria residing in the body, as direct or indirect interaction between humans and meat animals can lead to zoonotic transmission of bacteria. ²	National Advocacy for Rational Use of Medicine – Government programs such as the Antimicrobial Resistance Containment Program (AMRCP) list the regulation of antibiotics in food distribution channels in collaboration with the Thai FDA as a priority item on the list⁴, meaning as the level of advocacy increases with the implementation of this plan, the levels of antimicrobial resistant bacteria due to excessive use of antibiotics in meat will decrease. Quality of Food Safety Practices Thai people consume raw meat dishes such as fermented pork, and in samples collected from Chiang Rai, 80% of dishes were seen to have antimicrobial resistant enterococci.⁵ As antimicrobial resistance in meat production is managed, further transmission as a result of poor food safety practices such as cross contamination may decrease.	Country & Society This determinant operates at the country and society level as meat production and distribution is an endeavor requiring national level cooperation from administrations such as the Thai FDA, as seen in the Antimicrobial Resistance Containment Program (AMRCP) ⁴ . Consumers are also distributed across the country. ^{1,5}	Medium There is a strong risk associated with contraction of antimicrobial resistant bacteria such as ESBL resistant E.coli, which can cause severe foodborne illness. ¹ However, this risk of contraction can be mitigated through preparation such as cooking, as techniques such as raw fermentation may prove insufficient. ⁵

National Advocacy for Rational Use of Medicine	The WHO defines Rational Use of Medicine as “using correct medications that are relevant to clinical requirements of the patients, in exact doses according to the individual needs, for sufficient duration of time and at the lowest cost to them.” ⁶ National advocacy can be defined as the extent to which national organizations are advocating for rational use of medicine for their citizens. This can be measured by the presence or lack thereof of programs such as the Thai RDU Service Plan ⁷ , AMRCP ⁴ , and NSP-AMR ⁸ .	55% of Thailand’s Nakhon Nayok Province sampling population reported that their first provider when they felt sick and wanted to take antibiotics was a hospital rather than community pharmacist or local clinic. ⁶ The implementation of the RDU Service Plan, which operates through hospitals, led to antibiotic prescribing rates for respiratory infections, a leading motivator of antimicrobial consumption ⁶ , to decrease at a significantly higher rate than prior to implementation. ⁷ This increased rationality leads to decreased antimicrobial resistance, as irrational use leads to increased AMR. ⁶	Awareness of AMR Risks Among Citizens One goal of the Thai National Strategic Plan for Antimicrobial Resistance (NSP-AMR) was to increase public knowledge of antibiotics and awareness on AMR by 20% by 2021.⁹ As the level of advocacy increases through implementation of NSP-AMR, we saw an improvement in AMR awareness from 2017 to 2019.¹⁰ Education by Hospitalists One goal of the Thai RDU Service Plan is to increase knowledge about drugs prior to dissemination¹¹. As healthcare professionals are a major source of information about AMR⁹, the implementation of the RDU Service Plan will increase education by hospitalists to the public.	Country & Society These courses of action are taken through national governmental structures such as the Thai Ministry of Public Health⁷ and the Thai FDA,⁴ indicating that this determinant operates at the Country & Society level.	High This is a highly important determinant because of it directly decreases the rates of irrational administration and consumption of antibiotics⁷, and the decrease of irrational antibiotic use directly decreases AMR.⁶
Awareness of AMR Risks Among Citizens	An individual citizen’s awareness of the risks of antimicrobial resistance is defined as their comprehension of the relationship between rationality of drug use and clinical outcomes. ⁹ This can be	In Thailand’s Nakhon Nayok Province, “Knowledge” and “Attitude” towards antibiotics are strongly associated with antibiotic use behavior in a multiple linear regression analysis, with the insinuation	National Advocacy for Rational Use of Medicine A large component of National Advocacy for Rational Use of Medicine is improving the understanding of AMR by the public, such as the NSP-AMR	Individual Though people may access awareness through higher SEF level channels, AMR awareness is a determinant that varies	High Increased individual awareness directly decreases rates of infection by antimicrobial resistant drugs by prompting

	measured through a series of questions that measure if a person understands what an antibiotic is, the relationship between viral disease and antibiotics, the correct clinical use of antibiotics, and the consequences of misuse.⁹	that increased knowledge and more receptive attitude towards responsible antibiotic use lead to more rational drug behavior.⁶ More rational drug behavior does, in turn, decrease the risk of AMR.⁶	plan, which aimed to increase it by 2021⁹, and its implementation had a positive effect on public AMR awareness between 2017-2019.¹⁰ Completion of Antibiotic Courses AMR Awareness increases a person's likelihood of rational drug use⁶, and the WHO definition of rational drug use includes the usage of drugs for a sufficient amount of time⁶. Therefore, increasing a person's awareness of AMR makes them more likely to consume their antibiotics for a sufficient duration of time.	person to person based on numerous demographic factors such as gender, age, and education⁹, as well as individual exposures such as education from health professionals.⁹	more rational drug use.⁶ This determinant is also highly improvable through other determinants at higher SEF levels, such as the NSP-AMR planning to improve public awareness of AMR by 20% by 2021.⁹
Over the Counter Sale of Antibiotics via Community Pharmacists	The over the counter sale of antibiotic drugs via community pharmacists is defined as the amount of unprescribed antibiotics accessed through informal purchase at community pharmacies. This can be measured by calculating the percentage of sales made without a prescription among all antibiotic sales made at community pharmacies.¹²	In Thailand's Nakhon Nayok Province, 98% of patients seeking antibiotics who are not prescribed antibiotics at a hospital will then seek a secondary source for antibiotics⁶, and 85% of those patients will visit a local clinic or drugstore.⁶ Since it is legal to sell antibiotics without a prescription in Thailand¹³, they may then access these drugs with the clinical recommendation not to consume them. This is	Awareness of AMR Risks Among Citizens Community pharmacies can often be the most cost-effective and sometimes the only avenue of healthcare available to Thai people¹³. By discouraging over the counter sales via improvement of practice among healthcare providers, there is an improvement in AMR Risk awareness, because healthcare professionals in general have been identified as a major	Neighborhood & City Unlike Ministry of Public Health hospitals, which are operated by national entities⁷, community pharmacies operate not at an organized, national level, but rather are directly involved with their community of customers¹².	High Among community pharmacies around the globe, Thailand has been shown to have more than 80% of all antibiotic sales were made without a prescription.¹² This is a highly changeable determinant, as it has been shown that strict implementation of laws restricting over the counter sale of

		irrational drug use, which increases the risk of AMR. ⁶	source of antibiotics dispensing and education⁹ Education by Hospitalists Many times, community pharmacies are accessed after hospitals.⁶ In hospitals, there is insufficient understanding among future nurses about the risks of AMR¹⁴. As healthcare professionals are a major source of information about AMR⁹, by improving education by hospitalists, we increase knowledge about drug use. Increased knowledge leads to more rational drug behavior,⁶ which decreases over the counter purchasing of antibiotics.		antibiotics have drastically decreased sales in Brazil, South Korea, and other countries ¹² .
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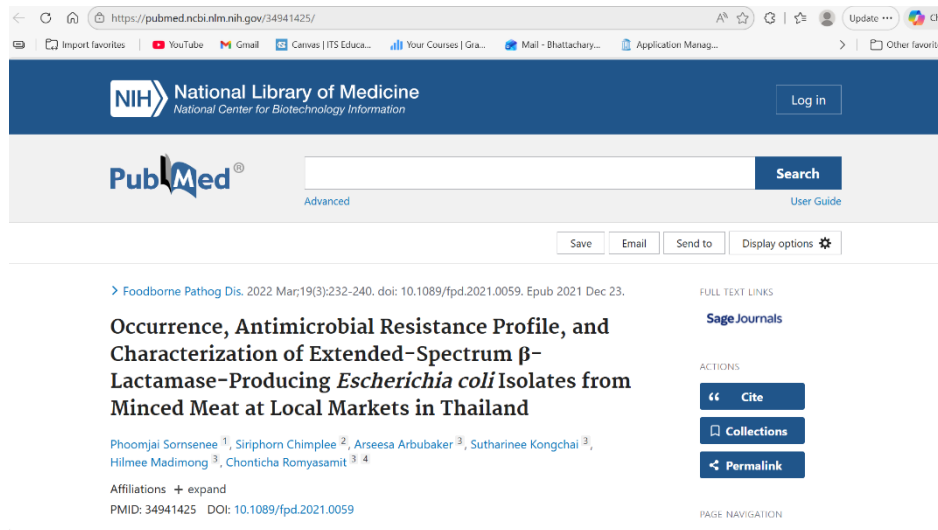
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REVIEW ARTICLE
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The public health issue of antibiotic residues in food and feed: Causes, consequences, and potential solutions

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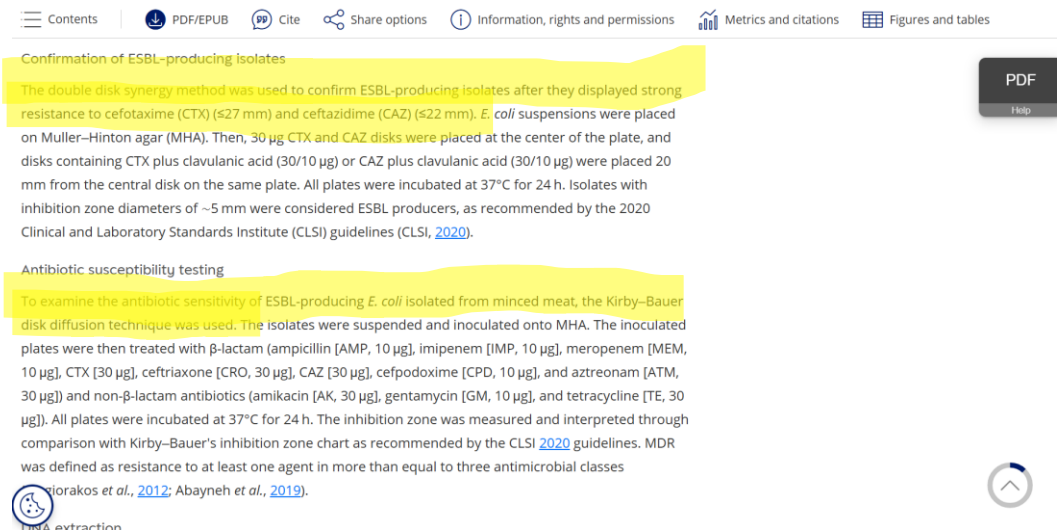
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Resistance to Antibiotics

The current transmission mechanisms of antibiotic resistance are better understood than those in the past. It has been established that the exposure of bacteria to low doses of antibiotics is likely to lead to bacterial adaptation, making them more resistant and more virulent [90-94]. Since antibiotic residues can be considered low or subtherapeutic doses, we assume that bacterial exposure can lead to adaptation. This mode of resistance acquisition is mainly based on spontaneous mutations and positive selection [95,96], alteration of the target with decreased affinity for the antibiotic [77,96], alternative metabolic pathways [79], change in membrane permeability, and efflux pumps [78,91]. The transmission of resistance between bacteria through a vertical (from one generation to another) or horizontal (transfer of resistance genes from one bacterium to another through conjugative plasmids, which play an essential role in the dissemination of resistance genes) gene transfer has worsened the situation [75], given that any direct or indirect interaction between humans and animals may lead to zoonotic transmission of antibiotic-resistant strains and genes from food animals to humans (Figure-1) [50]. Therefore, antibiotics should be carefully used to reduce the risks of resistance development and their complications in the management of

(3) prohibition of antibiotics whose toxicity is established (i.e., chloramphenicol, furazolidone, nitrofurazone, sulfonamides, and fluoroquinolones) and those more likely to induce direct or cross-resistance to antibiotics, (4) mandatory quality controls of food before marketing and the fight against black market products, and (5) permanent awareness of the dangers posed by antibiotic residues in food and resistance to antibiotics. Similarly, producers must (1) use antibiotics only when necessary and under the prescription of a veterinarian, (2) respect withdrawal times and other good practices related to antibiotics, (3) systematically test the presence of residues of the antibiotics used in their production, and (4) educate themselves on the regulations in force and respect them. In a similar manner, economic and straightforward field tests should be developed to identify antibiotic residues in animal products quickly. Meanwhile, methods such as heat treatment, activated charcoal, resin, and UV irradiation, may help inactivate antibiotics. Finally, probiotics and active phytochemical compounds should be further investigated and used as alternatives to antibiotics.

Conclusion

The residues of antibiotics in food are of severe public health concern. Although necessary, even essential in agriculture, they should nevertheless be



Review

Antimicrobial Resistance in Bacteria from Meat and Meat Products: A One Health Perspective

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Abstract: According to the 2030 Agenda of the United Nations, one of the sustainable development goals is to ensure sustainable consumption and production patterns. The need to ensure food safety includes, other than microbiological hazards, concerns with antimicrobial-resistant (AMR) bacteria. The emergence of resistant bacteria in the food industry is essentially due to the abusive, and sometimes incorrect, administration of antimicrobials. Although not allowed in Europe, antimicrobials are often administered to promote animal growth. Each time antimicrobials are used, a selective pressure is applied to AMR bacteria. Moreover, AMR genes can be transmitted to humans through

in food.

Antimicrobial resistance is considered a complex problem and a global health concern for both humans and animals. Around 2.8 million human cases of infections by antimicrobial-resistant bacteria and 700,000 deaths are reported annually, and this number could reach 10 million by 2050 if AMR is not reduced [16,17]. Antimicrobials have been used extensively and without respecting the therapy guidelines, mostly in low- and middle-income countries [18]. Resistant bacteria have been found in humans, animals, food, and the environment, leading to the transmission of resistance genes across bacterial species and between animals, humans, and the environment [18–20]. Bacteria isolated from food-producing animals have shown high AMR for most antimicrobials used in human medicine. About 54% of *Escherichia coli* and *Klebsiella* spp. showed high resistance to third-generation cephalosporine [16,18]. The existence of AMR bacteria in foods is mainly due to the excessive use of antimicrobials in food-producing animals and, consequently, the transmission of the AMR bacteria to humans through consumption (also known as “farm-to-fork” transmission) [21,22].

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3.5. Action 5: regulate the use and distribution of antibiotics in humans and food animals

Consumption of antibiotics that are dispensed by retail shops and drug stores is a driving force for developing AMR. Therefore, the Thailand Food and Drug Administration (FDA) appointed a subcommittee to reclassify the status and distribution channels of antibiotics to be used in humans and in food animals in 2013.

3.1. Detection of *Enterococcus* species

Among the 120 fermented pork samples, enterococci were detected in 96 samples (80.0%). A total of 119 enterococci were isolated, among which four species of *Enterococcus* were identified. The most prevalent species was *E. faecalis* ($n=82$; 68.9%), followed by *E. hirae* ($n=19$; 16.0%), *E. faecium* ($n=16$; 13.4%) and *E. gallinarum* ($n=2$; 1.7%). *E. casseliflavus* was not found in this study.

6.

The Predictors Influencing the Rational Use of Antibiotics Among Public Sector: A Community-Based Survey in Thailand

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Factors Related to Rational Behavior of Antibiotic Use

When performing MLR analysis, Enter method was firstly used for independent factors like Knowledge and Attitude because they are important variables that used to determine antibiotic use behavior. Forward Likelihood (LR) method was then performed as the best fit model for other variables to identify the predictors of rational behavior of antibiotic use. In MLR analysis by adjusting confounding factors, “Knowledge” and “Attitude” were significantly associated with antibiotic use behaviour, with the suggestion that respondents with sufficient knowledge and appropriate attitude were more likely to practice rational behavior. Both “Prevalence of antibiotic use within last 3 months” and “Visiting drugstore” were also found to be significant factors, with the respondents who did not use any kinds of antibiotics during last 3 months and visit drugstore if they had 3 main diseases, tended to show better practices than their counterparts. On the issue of age groups, the respondents between the ages of 35–56 had more rational use of antibiotics behavior than those with aged between 13 and 34 while older adults (57–78 years) were significantly associated with irrational behavior of antibiotics use comparing with the participants who were less than 34 years of age. For these results, age groups was related to rational drug use as concluded in Table 4.

7.

> Pharm Pract (Granada). 2021 Jan-Mar;19(1):2201. doi: 10.18549/PharmPract.2021.1.2201. Epub 2021 Feb 9.

Effects of a national policy advocating rational drug use on decreases in outpatient antibiotic prescribing rates in Thailand

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PMID: 33628347 PMCID: PMC7886315 DOI: 10.18549/PharmPract.2021.1.2201

Abstract

Objective: This study examined the effects of a national policy advocating rational drug use (RDU), namely, the 'RDU Service Plan', starting in fiscal year 2017 and implemented by the Thai Ministry of

Aiming for improved quality in the use of medicines, the features of the RDU Hospital Project were adopted by the MOPH and fully integrated into the Ministry's service plan policies at the beginning of fiscal year 2017 (October 2016). The project was then renamed the “RDU Service Plan”, and its implementation as a ministerial policy was expanded to the remaining MOPH hospitals nationwide that did not participate in the RDU Hospital Project. Hospital directors adopted the RDU Service Plan policies through their leading roles in establishing hospital strategies to achieve the RDU goals, encouraging hospital staff to conduct clinical practice according to the policy recommendations, and developing an internal monitoring and evaluation system. The RDU

Short Communication

Implementing national strategies on antimicrobial resistance in Thailand: potential challenges and solutions

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Highlights

- To contain antimicrobial resistance (AMR), the national strategic plan frames effective collaboration and

Goals

1. 50% reduction in AMR morbidity
2. 20% reduction in antimicrobial consumption in humans
3. 30% reduction in antimicrobial consumption in animals
4. 20% increase in public knowledge on AMR and awareness of appropriate use of antimicrobials
5. Capacity of the national AMR management system is increased to level 4 as measured by the WHO's Joint External Evaluation Tool for International Health Regulations (2005)

9.

RESEARCH ARTICLE

Knowledge and use of antibiotics in Thailand: A 2017 national household survey

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Abstract

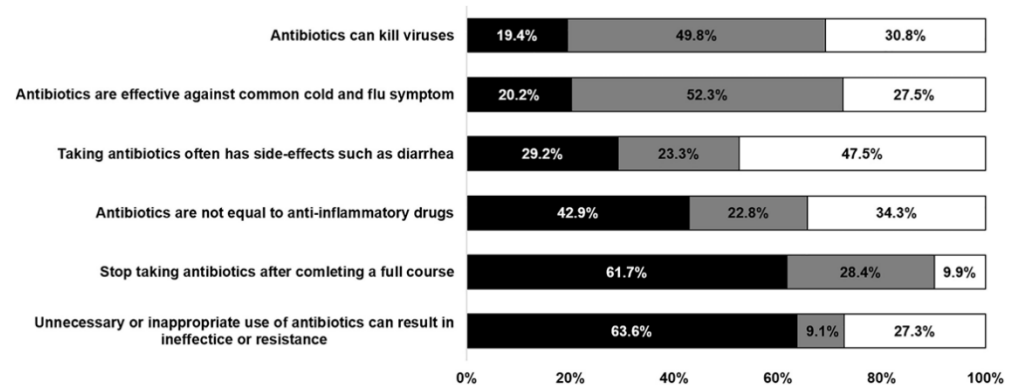
Background

The Thailand National Strategic Plan on Antimicrobial Resistance (AMR) 2017–2021, endorsed by the Thai Cabinet in 2016, aims to increase public knowledge about antibiotics and AMR awareness by 20% by 2021. This study assesses the prevalence of antibiotics use, clinical indications and sources; knowledge and access to information related to antibiotics and AMR; and factors related to level of knowledge and access to information among Thai adult population.

OPEN ACCESS

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> BMC Public Health. 2021 Nov 29;21(1):2188. doi: 10.1186/s12889-021-12237-y.

Population knowledge and awareness of antibiotic use and antimicrobial resistance: results from national household survey 2019 and changes from 2017

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Abstract

Background: Lack of knowledge and awareness on antimicrobial resistance (AMR) can result in irrational use of antibiotics, which is one of the major drivers of AMR. One goal of the Thailand

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Results: One month prior to the survey, 6.3% of population reported use of antibiotics (reduced from 7.9% to 2017), of which 98.1% received antibiotics through healthcare professionals and almost half (43.2%) for flu symptoms. During the last 12 months, 21.5% of Thai adults received information on the appropriate use of antibiotics and AMR (increased from 17.8% to 2017); mostly through health professionals (82.7%). On knowledge, 24.3% of adults gave correct answers to more than three out of six statements (three true and three false statements) (increased from 23.7% to 2017). The overall mean score of awareness of appropriate antibiotic use and AMR is 3.3 out of total score of 5.

Conclusions: Although progress was made on knowledge and awareness between 2017 and 2019, certain practices, such as use of antibiotics for flu symptoms and receiving information about antibiotic use and AMR, are inappropriate and inadequate. These findings require significant action,

11.

also include the importance of networking of which results showed that more members of well-established network submitted the key performance results at the end of the project; and (4) Collaboration at multiple levels were established to develop a single and unique patient information drug leaflet in the country, that will be linked to Mobile Application 'RDU รู้เรื่องยา' to improve the nation's health literacy in medication for Thai people. Major impacts of the research project are (1) its utilization as the development for the 15th Ministry of Public Health Service Plan started in the budget year 2017, (2) its acceptance as one of the tools being used for the healthcare accreditation process (the subjects of pharmaceutical and



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Date: 2560-09

Abstract

The objectives of Research Project in Development of Rational Drug Use (RDU) Hospital include: (1) to support the hospitals which join the RDU Hospital Project to achieve its goal by having a plan for concrete and effective actions based on the key advice P-L-E-A-S-E, (2) to strengthen the network of hospitals within the same geographical area, and (3) to support central tools in the implementation process of the hospital and network. The participatory action research (PAR) has been used to emphasize participation and action of local personnel and administrative team in 6 networks in the northern, northeastern-upper and lower, southern, central-eastern areas and the University Hospital Network (UHosNet). The suitable approaches are analyzed through the self-assessment of the hospital, network, and the performance indicators at the end of the project. Good practice and advice will be shared with the collaborators, stakeholders and hospitals continuing the RDU Hospital Development along with the central supporting tools. The key findings of the projects conducted within the two-year period (10/2015 to 9/2017) can be summarized as follows: (1) In cooperation with the hospitals and networks, several good actions were recognized in each aspect of the key advice P-L-E-A-S-E, which enable us to collect, recommend and distribute the report as a KM book; (2) Results of the performance indicators revealed improvement of the RDU Hospital processes and outputs in most hospitals, even if the targets are not reached in some metrics, thus imply the project success; (3) The key success factors are identified as the support of the executive team, multidisciplinary joint collaboration and the clear national policy. These also include the importance of networking of which results showed that more members of well-established network submitted the key performance results at the end of the project; and (4) Collaboration at multiple levels were established to develop a single and unique patient information drug leaflet in the country, that will be linked to Mobile Application 'RDU รู้เรื่องยา' to improve the nation's health literacy in medication for Thai people.

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Global access to antibiotics without prescription in community pharmacies: A systematic review and meta-analysis

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Abstract

Objective: To estimate the proportion of over-the-counter antibiotic requests or consultations that resulted in non-prescription supply of antibiotics in community pharmacies globally.

Methods: We systematically searched EMBASE, Medline and CINAHL databases for studies published from January 2000 to September 2017 reporting the frequency of non-prescription sale and supply of antibiotics in community pharmacies across the world. Additional articles were identified by checking reference lists and a Google Scholar search. A random effects meta-analysis was conducted to

Strict implementation of restrictions on over-the-counter sales of antibiotics has been shown to be effective in reducing non-prescription antibiotic consumption in Brazil, Chile and South Korea and Mexico.³²⁻³⁴ Given that many countries have laws prohibiting over-the-counter sales of antibiotics, there is a need to ensure that these laws are strictly enforced. However, it

Article

Antibiotics' Use in Thailand: Community Pharmacists' Knowledge, Attitudes and Practices

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Abstract: Thailand spends \$203 million on antibiotics every year, and patients can still access

Table 5. Correlation of Knowledge, Attitude, and Practice scores.

	Spearman's Rho	<i>p</i> -Value
Knowledge–Attitude	0.100	0.053
Attitude–Practice	0.149	0.004 ***
Knowledge–Practice	0.071	0.174

*** = significant at *p* value < 0.01 level.

2.2.5. Relationship between KAP Scores and Community Pharmacists' Demographics

The knowledge, attitude, and practice scores were compared in relation to the participants' demographics to explore the differences across gender, age, education, professional

14.